

ChargeHub – Smart EV Charging & Trip Planner

SoftwareRequirementsSpecification(SRS)Document • Version1.0

STUDENT NAME

ROLL NUMBER

SECTION / GROUP

APPROVED TECH STACK

HTML5,CSS3,JS,Figma, GitHub

1. Executive Summary & Purpose

ChargeHub is an intelligent electric vehicle (EV) charging station finder, real-time availability tracker, slot reservation tool, and route-planning application. Designed as an academic software engineering project, this document provides complete implementation specifications, architectural boundaries, and software requirements suitable for faculty evaluation, frontend development, and future database integration.

2. Project Scope & Target Objectives

The primary objectives of the ChargeHub client-side application include:

- **Station Locating:** Interactive listing and map-based rendering of charging stations.
- **Slot Reservation:** Booking mechanism with time slot selection saved via LocalStorage.
- **Trip & Range Planner:** Route planning with battery consumption estimations and charging stop recommendations.
- **Cost Calculation:** Automatic charging cost estimation based on required kWh and tariff rates.

3. Approved Technology Stack & Constraints

To ensure a realistic academic scope while maintaining a professional system, the stack is restricted to:

HTML5

CSS3 (Flexbox/Grid)

Vanilla JavaScript (ES6+)

Figma (UI/UX)

GitHub & Pages

Constraint Note: In accordance with project rules, no external backend frameworks (Node/Python) or heavy JS frameworks (React/Angular) are used in Phase 1. State persistence is handled natively via browser LocalStorage and pre-calibrated JSON datasets.

4. Functional Requirements (FR)

Functional requirements define the core operational features that the ChargeHub application must provide to users.

REQ ID	MODULE NAME	REQUIREMENT DESCRIPTION
FR-01	Station Locator	The system shall display available EV stations filtered by location, connector type (Type 2, CCS, CHAdeMO), and power rating (kW).
FR-02	Status Tracking	The system shall visually indicate charger status (Available, Occupied, Under Maintenance) with real-time UI updates.
FR-03	Slot Reservation	Users shall select a station, connector, and time slot to generate a booking confirmation stored in localStorage.
FR-04	Route Planner	Users shall enter start point, destination, current battery %, and vehicle range to receive charging stop recommendations.
FR-05	Cost Estimator	The system shall calculate estimated charging costs based on tariff rates (per kWh) and required battery charge boost.
FR-06	Booking History	Users shall be able to view, manage, or cancel active slot reservations from a dedicated profile dashboard.

5. Module Architecture & UI Components

1. Station Search & Filter Module Interactive search bar with dynamicJavascript filtering by distance, charger type, and speed. Renders station cards dynamically into the DOM.	2. Interactive Slot Booking Modal Pop-upmodal UIfor selectingcharging hours. Validates time overlaps before writing reservation JSON into browser storage.
3. Smart Route & Trip Calculator Algorithm thatcalculatesremainingbattery distance and pinpoints mandatory charging station stops along the route trajectory.	4. User Dashboard & Digital Pass Rendersan interactivedigital bookingslip with QR placeholder, station location details, and active status badge.

6. External Interface Requirements

- User Interface (UI):** Responsive web interface styled using a custom Blue Theme CSS Palette (`#1e3a8a`, `#2563eb`, `#eff6ff`). High-fidelity prototypes hosted on Figma.
- Software Interfaces:** HTML5 Web Storage API (`localStorage`) for data persistence; DOM API for dynamic DOM manipulations; standard Fetch API for mock static JSON dataset loading.
- Hardware Interfaces:** Compatible with touch-based mobile viewports (minimum 375px width) and desktop monitors (1080p+).

7. Non-Functional Requirements (NFR)

CATEGORY	METRIC / TARGET	SPECIFICATION DETAIL
Performance	Load Time < 1.5s	All local assets (CSS/JS) must load rapidly without external heavy dependencies. UI filters render in <100ms.
Usability	Responsive Layout	Mobile-first fluid responsive grid adaptively formatted for desktop, tablet, and smartphone screens.
Design Theme	Blue Theme Standard	Consistent color system leveraging Deep Navy (`#0f172a`), Royal Blue (`#2563eb`), and Sky Blue (`#93c5fd`).
Reliability	Session Persistence	Booking state and user preferences must survive browser tab reloads through synchronous localStorage handling.
Maintainability	Clean Code Base	Code organized in modular JS files, semantic HTML5 tags, and version-controlled systematically on GitHub.

8. Project Workflow & Software Lifecycle

Phase 1: Figma Prototyping

Design full wireframes, user flow diagrams, and blue-theme UI components in Figma. Validate user experience navigation.

Phase 2: HTML/CSS Frontend

Develop semantic, accessible HTML templates and clean CSS styles using modern Flexbox and Grid layouts.

Phase 3: JavaScript Integration

Implement dynamic filtering, station listing algorithms, trip calculations, and slot reservation logic using LocalStorage.

Phase 4: Testing & GitHub Deployment

Execute cross-browser UI testing, validate reservation edge cases, and publish live on GitHub Pages.

9. Faculty Evaluation & Verification Matrix

This SRS serves as the official design benchmark for faculty review. Evaluation criteria are mapped directly to implementation deliverables:

- **UI/UX Fidelity:** Verification of Figma prototype implementation into code.
- **DOM Logic & Algorithms:** Assessment of pure JavaScript filtering and cost estimation math.
- **Documentation Quality:** Evaluation against standard IEEE Software Requirements Specification structure.
- **Repository Management:** Review of commit history, clean file structure, and GitHub deployment.